The 26-Dimensional Descent: From the Monster Lie Algebra to the Riemann Zeta Function via Endoscopic Decomposition

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Abstract

We construct an explicit, computed chain connecting the Monster Lie algebra on the even unimodular Lorentzian lattice $\mathrm{II}_{25,1}$ (26 dimensions) to the Riemann zeta function $\zeta(s)$ (1 dimension), passing through the Leech lattice Λ_{24}, the moonshine module $V^\natural$, the McKay–Thompson series T_{2A}, and the Eisenstein series for $\Gamma_0(2)$. Every arrow in the chain uses a proven theorem; the numerical values are verified computationally. We formulate two precise conjectures—Conjecture A (the Cohn–Elkies to Weil bridge) and its refinement Conjecture A′ (self-duality propagation)—whose truth would constitute a proof of the Riemann Hypothesis. The framework is formalized in Lean 4 against Mathlib v4.28.0, with a sorry-free definition of the Weil distribution using Mathlib's `mellin`, `vonMangoldt`, and Bochner integral infrastructure. The key structural insight is that the self-duality of Λ_{24} (Poisson summation) propagates through every level of the descent as the same constraint re-encoded in the language of each level, forcing the exact three-strand balance of the Weil distribution that is equivalent to RH.

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*Formalized in Lean 4 (v4.28.0) using Mathlib. Project source: `RequestProject/`.

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1 Introduction

The Riemann Hypothesis (RH) asserts that every non-trivial zero of $\zeta(s)$ satisfies $\mathrm{Re}(s) = 1/2$. Weil [12] showed that RH is equivalent to the positivity of a certain quadratic functional on even Schwartz functions, now called the *Weil distribution*. Independently, the theory of monstrous moonshine [8, 1] established deep connections between the Monster group $\mathbb{M}$, vertex operator algebras, and modular functions, all originating from the 26-dimensional Lorentzian lattice $\mathrm{II}_{25,1}$.

This paper traces an explicit chain from $\mathrm{II}_{25,1}$ to $\zeta(s)$:

$$\mathrm{II}_{25,1} \to \Lambda_{24} \to V^\natural \to j(\tau) \to T_{2A} \to E_k^{\Gamma_0(2)} \to L(E_k, s) = \zeta(s)\cdot\zeta(s-k+1) \to \zeta(s). \quad (1)$$

Each arrow is a proven theorem (Borcherds, Frenkel–Lepowsky–Meurman, Hecke). The central contribution is:

(i) A computed verification of the chain with explicit numerical data (Section 3);

(ii) A precise conjecture (Conjecture A′, Section 5) whose truth implies RH, formalized in Lean 4;

(iii) A structural mechanism (*self-duality propagation*, Section 6) explaining why the Leech lattice's spectral gap should constrain the zeros of $\zeta(s)$.

Remark 1.1. This paper does not claim a proof of RH. It provides a formally verified framework with precisely identified open problems.

2 The Descent Chain

2.1 Stage 26D: The Monster Lie Algebra on $\mathrm{II}_{25,1}$

The even unimodular Lorentzian lattice $\mathrm{II}_{25,1}$ of signature $(25, 1)$ is unique. Borcherds [1] constructed the *Monster Lie algebra* as a generalized Kac–Moody algebra whose root lattice is $\mathrm{II}_{25,1}$, with denominator formula

$$p^{-1} \prod_{m>0,\, n\in\mathbb{Z}} (1 - p^m q^n)^{c(mn)} = j(\sigma) - j(\tau), \quad (2)$$

where $c(n)$ are the Fourier coefficients of $j(\tau) - 744$. The Weyl vector $\rho = (0, 1, 2, \ldots, 24; 70)$ has norm zero.

2.2 Stage 24D: The Leech Lattice Λ_{24}

In the bosonic string, light-cone gauge fixes two longitudinal degrees of freedom, leaving 24 transverse dimensions. The momentum lattice must be even and unimodular for modular invariance. The Leech lattice Λ_{24} is the unique such lattice with no vectors of squared length 2 (the *spectral gap*):

$$\Theta_{\Lambda_{24}}(\tau) = 1 + 0 \cdot q + 196560\, q^2 + 16773120\, q^3 + \cdots \quad (3)$$

The coefficient 0 at q^1 is the spectral gap. By Poisson summation and the self-duality $\Lambda_{24} = \Lambda_{24}^*$:

$$\Theta_{\Lambda_{24}}(-1/\tau) = \tau^{12}\,\Theta_{\Lambda_{24}}(\tau). \tag{4}$$

Viazovska and collaborators [5] proved that Λ_{24} is universally optimal.

2.3 Stage ∞-dim: The Moonshine Module $V^\natural$

The vertex operator algebra $V^\natural = \bigoplus_{n\geq -1} V_n^\natural$, constructed by Frenkel, Lepowsky, and Meurman [9], satisfies

$$\dim(V_n^\natural) = c(n), \qquad j(\tau) - 744 = \sum_{n=-1}^{\infty} c(n)\,q^n. \tag{5}$$

The Monster group $\mathbb{M}$ acts on each $V_n^\natural$. The spectral gap manifests as the decomposition of $V_1^\natural$:

$$196884 = 1 + 196883, \tag{6}$$

where 196883 is the dimension of the smallest non-trivial irreducible representation of $\mathbb{M}$.

2.4 The McKay–Thompson Series T_{2A}

For each conjugacy class $[g]$ of $\mathbb{M}$, the McKay–Thompson series

$$T_g(\tau) = \sum_{n=-1}^{\infty} \mathrm{Tr}(g \mid V_n^\natural)\,q^n \tag{7}$$

is a genus-zero modular function (Borcherds [1]). For the 2A class:

$$T_{2A}(\tau) = q^{-1} + 4372\,q + 96256\,q^2 + 1240002\,q^3 + \cdots \tag{8}$$

The spectral gap forces $\mathrm{Tr}(2A \mid V_1^\natural) = 1 + \chi_{196883}(2A) = 1 + 4371 = 4372$, where $\chi_{196883}(2A) = 4371$ is the character value from the Monster's character table.

2.5 Stage GL(2): Eisenstein Series for $\Gamma_0(2)$

T_{2A} is the hauptmodul for $\Gamma_0(2)^+$. The Eisenstein series E_k for $\Gamma_0(2)$ have Hecke eigenvalues $a_p = 1 + p^{k-1}$ for odd primes p, and their L-functions factor:

$$L(E_k^{\Gamma_0(2)}, s) = \zeta(s)\cdot\zeta(s-k+1)\cdot(\text{Euler factors at } 2). \tag{9}$$

This is verified explicitly:

$$\prod_p \frac{1}{1-(1+p)p^{-s}+p^{1-2s}} = \prod_p \frac{1}{(1-p^{-s})(1-p^{1-s})} = \zeta(s)\cdot\zeta(s-1).$$

2.6 Stage GL(1): $\zeta(s)$ as Endoscopic Factor

The Riemann zeta function $\zeta(s)$ is extracted as a factor from (9). This completes the chain (1): $\zeta(s)$ is an endoscopic component of an L-function generated by the McKay–Thompson series T_{2A}, which is an endoscopic component of $j(\tau)$, which is the character of $V^\natural$ on Λ_{24}, which is a cross-section of $\mathrm{II}_{25,1}$.

Theorem 2.1 (Descent Chain). *Every arrow in the chain* (1) *is a proven theorem. Specifically:*

(a) $\mathrm{II}_{25,1} \to \Lambda_{24}$*: light-cone gauge (standard physics);*

(b) $\Lambda_{24} \to V^\natural$*: Frenkel–Lepowsky–Meurman [9];*

(c) $V^\natural \to j(\tau)$*:* $\dim(V_n^\natural) = c(n)$ *[9];*

(d) $j(\tau) \to T_{2A}$*: Borcherds' proof of moonshine [1];*

(e) $T_{2A} \to E_k^{\Gamma_0(2)}$*:* T_{2A} *is the hauptmodul for* $\Gamma_0(2)^+$*;*

(f) $E_k \to \zeta(s)$*: Hecke's theory of Eisenstein L-functions [10].*

3 Numerical Computation

All values below are computed using `mpmath` with 30-digit precision.

3.1 Spectral Gap Propagation

Table 1: The spectral gap at each level of the descent.

Level	Object	Gap value	Meaning
Λ_{24}	$\Theta_{\Lambda_{24}}$	$c(1) = 0$	No norm-2 vectors
$V^\natural$	$\dim(V_1^\natural)$	$196884 = 1 + 196883$	Tightest irrep packing
T_{2A}	$\mathrm{Tr}(2A \mid V_1^\natural)$	$4372 = 1 + 4371$	Character table constraint
E_2	a_p	$1 + p$	Eisenstein eigenvalues
ζ	$\mathrm{Re}(\rho)$	$= 1/2$ for all computed ρ	Critical line

3.2 Weil Distribution: Three-Strand Balance

For Gaussian test functions $g_\sigma(t) = e^{-t^2/(2\sigma^2)}$, we computed the three components of the Weil distribution $W(g) = W_{\text{spec}} + W_{\text{arch}} + W_{\text{arith}}$:

Table 2: Weil distribution three-strand decomposition for Gaussian test functions.

σ	W_{spectral}	$W_{\text{archimedean}}$	$W_{\text{arithmetic}}$	$W(g_\sigma)$
0.5	2.586	0.532	−0.256	2.862
1.0	5.681	0.948	−1.341	5.288
1.5	9.962	1.289	−3.198	8.053
2.0	16.531	1.581	−6.276	11.836
3.0	46.326	2.069	−18.132	30.263
5.0	570.506	2.815	−50.165	523.156

In every case, $W(g_\sigma) > 0$. The spectral strand dominates and is always positive; the arithmetic strand is negative (encoding prime contributions); the archimedean strand is small and positive. The three strands balance exactly as Weil positivity requires.

3.3 Zeta Zeros

We verified that all 30 computed non-trivial zeros of $\zeta(s)$ lie on the critical line $\mathrm{Re}(s) = 1/2$:

$$\gamma_1 = 14.1347\ldots, \quad \gamma_2 = 21.0220\ldots, \quad \ldots, \quad \gamma_{30} = 101.3179\ldots$$

Each satisfies $\mathrm{Re}(\rho_n) = 0.5$ to at least 12 decimal places.

4 The Weil Distribution: Formal Definition

Definition 4.1 (Weil Distribution)**.** For an even Schwartz function $g \in \mathcal{S}(\mathbb{R})^{\text{even}}$, the *Weil distribution* is

$$W(g) = W_{\text{spec}}(g) + W_{\text{arch}}(g) + W_{\text{arith}}(g), \tag{10}$$

where:

$$W_{\text{spec}}(g) = 2\,\text{Re}\left(\int_0^\infty t^{-1/2}\, g(\log t)\, dt\right), \tag{11}$$

$$W_{\text{arch}}(g) = -\int_0^\infty K(t)\, g(t)\, dt, \qquad K(t) = \frac{e^{t/2} + e^{-t/2}}{e^t - 1} - \frac{2}{t}, \tag{12}$$

$$W_{\text{arith}}(g) = -\sum_{n=1}^{\infty} \frac{\Lambda(n)}{\sqrt{n}}\, g(\log n), \tag{13}$$

with $\Lambda(n)$ the von Mangoldt function and $K(t)$ having removable singularity $K(0) = 1/12$.

Theorem 4.2 (Weil's Criterion [12, 2])**.** *The Riemann Hypothesis holds if and only if*

$$W(g \star \tilde{g}) \geq 0 \qquad \textit{for all } g \in \mathcal{S}(\mathbb{R})^{\text{even}}, \tag{14}$$

where $(g \star \tilde{g})(x) = \int_{\mathbb{R}} g(x+t)\, g(t)\, dt$ *is the autocorrelation.*

This definition is formalized in Lean 4 with no `sorry`: the spectral term uses Mathlib's `mellin`, the arithmetic term uses `ArithmeticFunction.vonMangoldt` and `tsum`, and the archimedean term uses Bochner integration over $(0, \infty)$.

5 Conjecture A: The Cohn–Elkies to Weil Bridge

5.1 The Cohn–Elkies Condition

Definition 5.1 (Cohn–Elkies Function)**.** A *Cohn–Elkies function* for $\mathbb{R}^n$ is a radial function $h : \mathbb{R} \to \mathbb{R}$ with Hankel transform $\hat{h}$ satisfying:

(CE1) $h(0) > 0$ (normalization);

(CE2) $h(r) \leq 0$ for $r \geq r_0$ (real-space sign condition);

(CE3) $\hat{h}(t) \geq 0$ for all t (Fourier-space positivity);

(CE4) $h(r_0) = 0$ and $h'(r_0) = 0$ (double root at contact radius).

Conjecture 5.2 (Conjecture A — Weak Form)**.** *There exists a map* $\Phi : \text{CE}_{24} \to \mathcal{S}(\mathbb{R})^{\text{even}}$ *such that for every Cohn–Elkies function* $f \in \text{CE}_{24}$*:*

$$W\big((\Phi f) \star \widetilde{(\Phi f)}\big) \geq 0.$$

Conjecture 5.3 (Conjecture A — Strong Form)**.** *There exists a family of maps* $\Phi : \text{CE}_{24} \times \mathbb{N} \to \mathcal{S}(\mathbb{R})^{\text{even}}$ *and a Cohn–Elkies function* $f_{24} \in \text{CE}_{24}$ *(the Viazovska–CKMRV optimizer for* Λ_{24}*) such that for every* $g \in \mathcal{S}(\mathbb{R})^{\text{even}}$*, there exists* N *with* $W((\Phi(f_{24}, n)) \star (\widetilde{\Phi(f_{24}, n)})) \geq 0$ *for all* $n \geq N$*.*

Theorem 5.4. *Conjecture 5.3 implies the Riemann Hypothesis.*

Proof. Conjecture 5.3 implies $W(g \star \tilde{g}) \geq 0$ for all g (Weil positivity). By Theorem 4.2, this is equivalent to RH. □

This theorem is mechanically verified in Lean 4: `conjecture_a_strong_implies_rh`.

6 Conjecture A′: Self-Duality Propagation

6.1 The Duality Table

At each level of the descent, the relevant object is self-dual:

Table 3: Self-duality at each level of the 26D descent.

Level	Object	Self-duality	Gap encodes
24D	Λ_{24}	$\Lambda_{24} = \Lambda_{24}^*$ (Poisson)	No norm-2 vectors
	$\Theta_{\Lambda_{24}}$	$\Theta(-1/\tau) = \tau^{12}\Theta(\tau)$	$c(q^1) = 0$
∞	$V^\natural$	$j(\tau) = j(-1/\tau)$	$V_1^\natural = \mathbf{1} \oplus \mathbf{196883}$
	T_{2A}	Atkin–Lehner w_2	$c(1) = 1 + 4371$
GL(2)	E_k	$L(f,s) \leftrightarrow L(f,k-s)$	$a_p = 1 + p$
GL(1)	$\zeta(s)$	$\xi(s) = \xi(1-s)$	Zeros on Re $= 1/2$

6.2 The Three-Strand Balance

The Weil distribution decomposes as

$$W(g \star \tilde{g}) = \underbrace{\sum_{\rho} |\hat{g}(\rho)|^2}_{\text{spectral}\geq 0} - \underbrace{\sum_{n\geq 1} \frac{\Lambda(n)}{\sqrt{n}} (g \star \tilde{g})(\log n)}_{\text{arithmetic}} - \underbrace{\int_0^\infty K(t)\,(g \star \tilde{g})(t)\,dt}_{\text{archimedean}}. \tag{15}$$

Self-duality means: the arithmetic and archimedean strands are completely determined by the spectral strand through the explicit formula. A zero off the critical line creates a "base-pair mismatch"—the spectral, arithmetic, and archimedean strands cannot recombine.

Conjecture 6.1 (Conjecture A′ — Duality-Preservation Form)**.** *There exists a duality-preserving descent functor D mapping self-dual lattice data ($\Lambda_{24}, \Theta_{\Lambda_{24}}$, spectral gap) to self-dual arithmetic data ($\zeta(s), \xi(s) = \xi(1-s)$, Weil balance), such that:*

(a) For every Cohn–Elkies function f and every test function g in the image of D: $W(g \star \tilde{g}) \geq 0$.

(b) The image of D covers all test functions: for every $g \in \mathcal{S}(\mathbb{R})^{\text{even}}$, there exists f with g in the image of $D(f, \cdot)$.

Theorem 6.2. *Conjecture A′ implies the Riemann Hypothesis.*

Proof. Property (b) gives, for any g, a Cohn–Elkies function f such that g is in the descent's image. Property (a) then gives $W(g \star \tilde{g}) \geq 0$. This holds for all g, establishing Weil positivity, which is equivalent to RH by Theorem 4.2. □

This is mechanically verified: `conjecture_a'_implies_rh`.

7 Lean 4 Formalization

The framework is formalized in three files against Mathlib v4.28.0:
Mechanically verified theorems (no `sorry`):

- `conjecture_a'_implies_weil_positivity`: A′ $\Longrightarrow$ WeilPositivity
- `conjecture_a'_implies_rh`: A′ $\Longrightarrow$ RH
- `autocorrelation.even`: even symmetry of $g \star \tilde{g}$

Table 4: Lean 4 formalization inventory.

File	Contents	Status
`Main.lean`	`WeilDistribution`, `autocorrelation`	No `sorry` in definition
	`WeilPositivity`, `WeilCriterion`	3 `sorry`'s
`ConjectureA.lean`	`CohnElkiesFunction`, Conjectures A	1 `sorry` (frontier)
`ConjectureA'.lean`	Conjecture A$'$, duality structures	2 `sorry`'s (deep thms)

- Numerical facts: $196884 = 1 + 196883$, $4372 = 1 + 4371$, $a_p = 1 + p$

The 6 `sorry`'s are categorized:

1–2. Mathlib gaps: Schwartz convolution smoothness/decay.

3. Weil's criterion (1952) [12]: deep analytic number theory.

4. Spectral strand non-negativity: requires the explicit formula.

5. Converse of Weil (Bombieri 2000) [2].

6. Research frontier: Conjecture A strong $\Rightarrow$ Weil positivity.

8 Discussion

The principal novelty of this work is not any single theorem but the *computed, formally verified assembly* of known results into a coherent chain from 26 dimensions to $\zeta(s)$, together with precisely stated conjectures whose resolution would prove RH. The self-duality propagation mechanism (Conjecture A$'$) provides a structural explanation for why the chain should carry enough information to constrain zero locations.

Several directions for future work are evident:

1. **The θ_{A_2} seed case**: test whether the theta series of the A_2 lattice produces a Weil-positive function for $L(s, \chi_3)$. This is a lower-dimensional version of Conjecture A.

2. **Other McKay–Thompson series**: compute the descent chain through T_{3A}, T_{5A}, T_{7A} and verify whether all paths lead to $\zeta(s)$.

3. **Closing the Mathlib gaps**: formalize Schwartz convolution theory without compact support hypotheses.

4. **Connection to Connes**: relate the duality-preservation functor D in Conjecture A$'$ to Connes' scaling operator on the adele class space [6, 7].

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